

Incident Response Report

WormLocker 2.0 Ransomware Analysis

Incident ID: 01

Date of Incident: 09/11/2025

Report Date: 15/11/2025

SOC Responsible: MADENE Meriem

Malware Analyst: MOSTEFAOUI Manel

Monitoring and Logging Responsible: BRAHMI Sarah Kaouther

Classification: Public Incident Reponse Report

Contents

1	Executive Summary	2
1.1	Key Findings	2
2	Phase 1: Preparation & Identification	3
2.1	Detection Method	3
2.2	Initial Indicators of Compromise (IOCs)	3
2.3	Artifacts Collected	3
3	Phase 2: Containment	6
3.1	Immediate Actions Taken	6
4	Phase 3: Eradication	7
4.1	Malware Removal Steps	7
4.2	File Recovery Attempts	7
5	Phase 4: Recovery	8
5.1	System Restoration	8
5.2	Validation Steps	8
5.3	Time to Recovery	9
6	Phase 5: Lessons Learned	10
6.1	Incident Summary	10
6.2	Root Cause Analysis	10
6.3	What Worked Well	10
6.4	What Could Be Improved	10
7	Recommendations	11
7.1	Immediate Actions	11
7.2	Long-Term Improvements	11
8	Appendices	12
8.1	Appendix A: Technical Details	12
8.2	Appendix B: IOC List	12
8.3	Appendix C: Tools Used	12
9	Report Sign-off	14

1 Executive Summary

Brief overview of the incident in 3-4 sentences.

[Describe what happened, impact, and outcome]

1.1 Key Findings

- **Malware Family:** WormLocker 2.0 Ransomware
- **Attack Vector:** Drive-by download
- **Systems Affected:** Windows 10 VM
- **Data Impact:** Screen lock, file encryption attempt

2 Phase 1: Preparation & Identification

2.1 Detection Method

How was the incident discovered?

The detection was found by The System Monitoring and Logging we made.

2.2 Initial Indicators of Compromise (IOCs)

- **Malware Filename:** WormLocker2.exe
- **File Path:** C:\Users\Yamina\Downloads
- **File Hash (SHA256):** d4c83205cf6f3098ab6a757312525f4d14a57a819306eeea5c0d022b00b38cf3
- **Suspicious Process:** WormLocker2.exe, conhost.exe, cmd.exe.
- **Network Activity:** No network activity detected.
- **Ransom Note:** ransom_note.vbs

2.3 Artifacts Collected

- Screenshot of ransom note
- Process Monitor logs
- Network traffic capture (PCAP)
- Memory dump
- Disk dump

Screenshots:

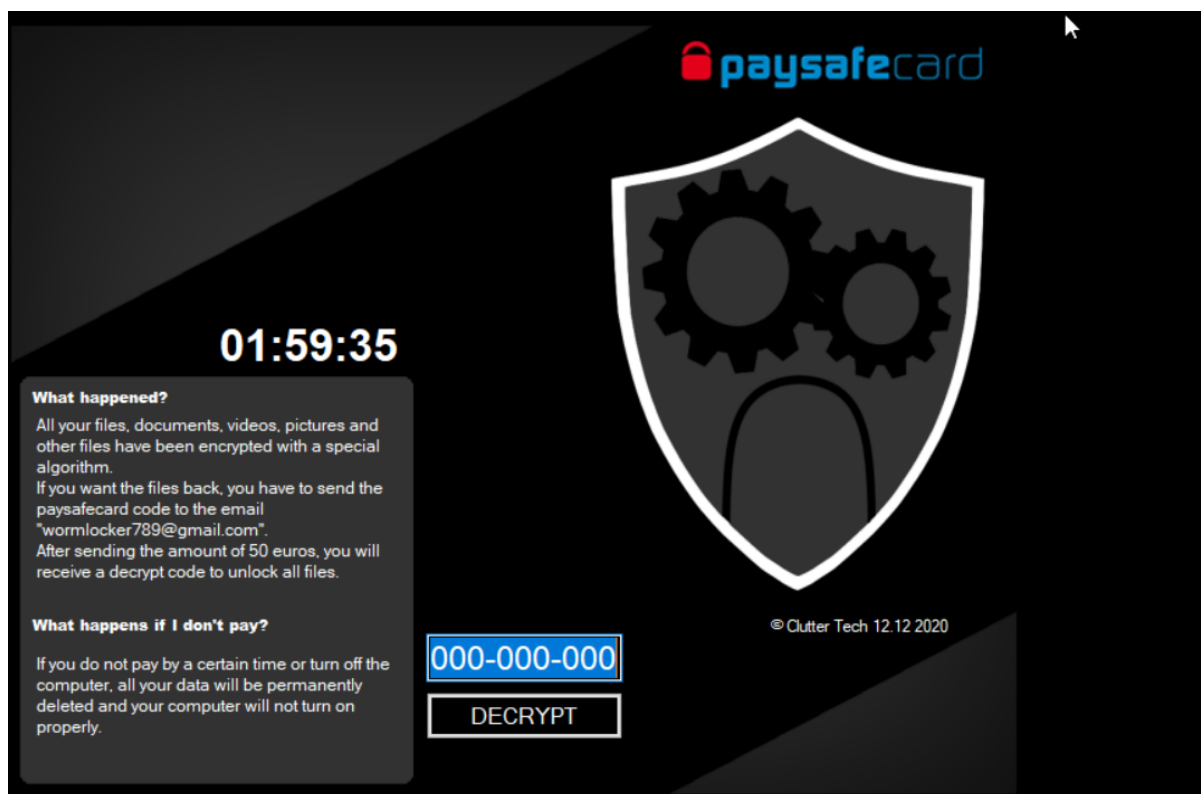


Figure 1: Extracted ransom note

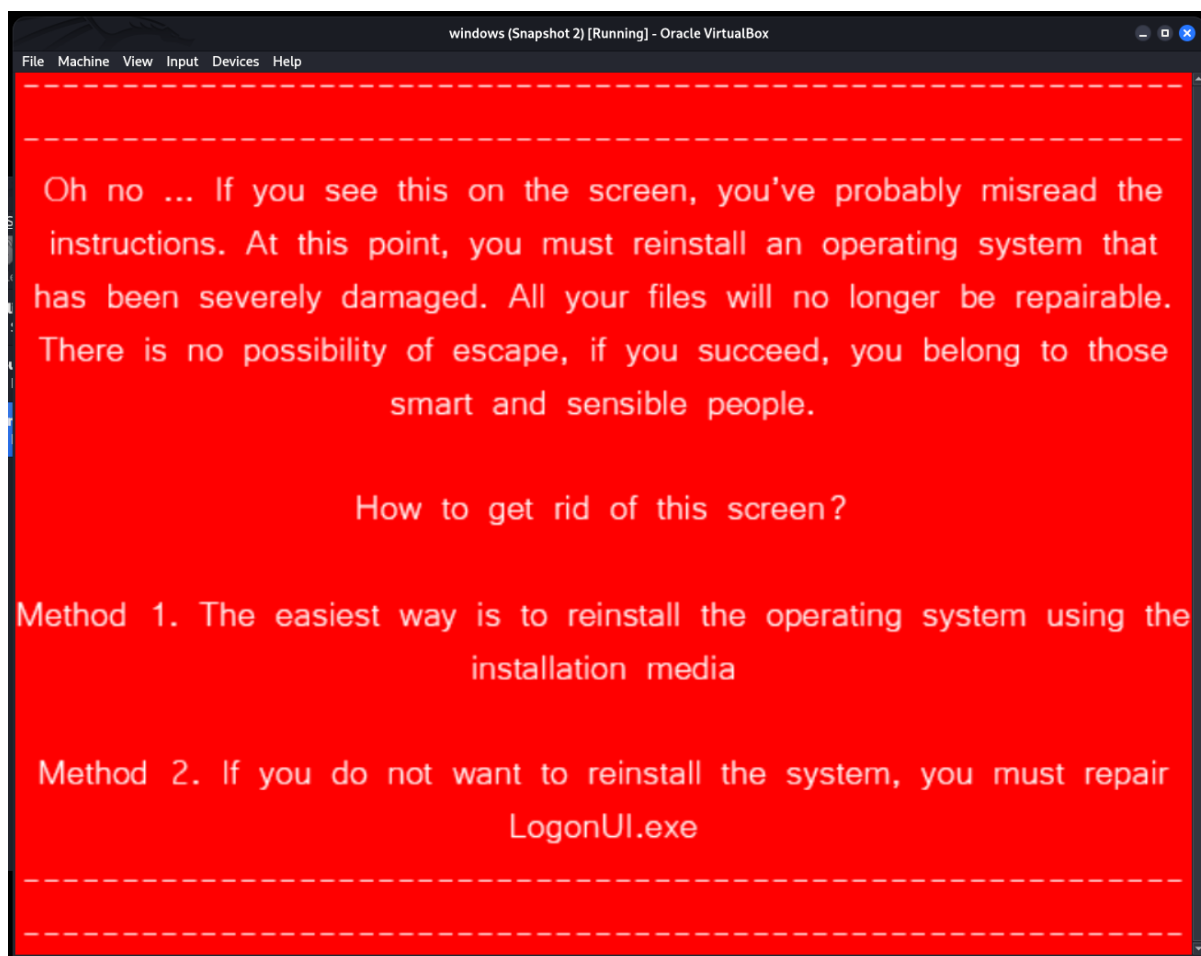


Figure 2: Extracted ransom note after rebooting the system

3 Phase 2: Containment

3.1 Immediate Actions Taken

Goal: Prevent further spread and damage

1. Network Isolation

- Disconnecting the VM immediately from the network at the time of the detection of the malware.

2. Process Termination

- Could not terminate the process because the screen was locked, task manager was disabled and there was no way to do any actions, we only saved the state of the machine for later investigation.

4 Phase 3: Eradication

4.1 Malware Removal Steps

1. Manual Deletion

- Manually deleting the file before executing it as soon as it gets detected by the IDS.
- Or waiting for the windows defender to delete the file after scanning it.

2. Antivirus Scan

- Windows Defender
- After the scan the file was automatically deleted.

3. Persistence Mechanisms

- Found the LogonUI.exe file, which is responsible for persistence and for crashing the system after rebooting the system.

4.2 File Recovery Attempts

- Files completely encrypted
- Tried some public keys that was found, none worked

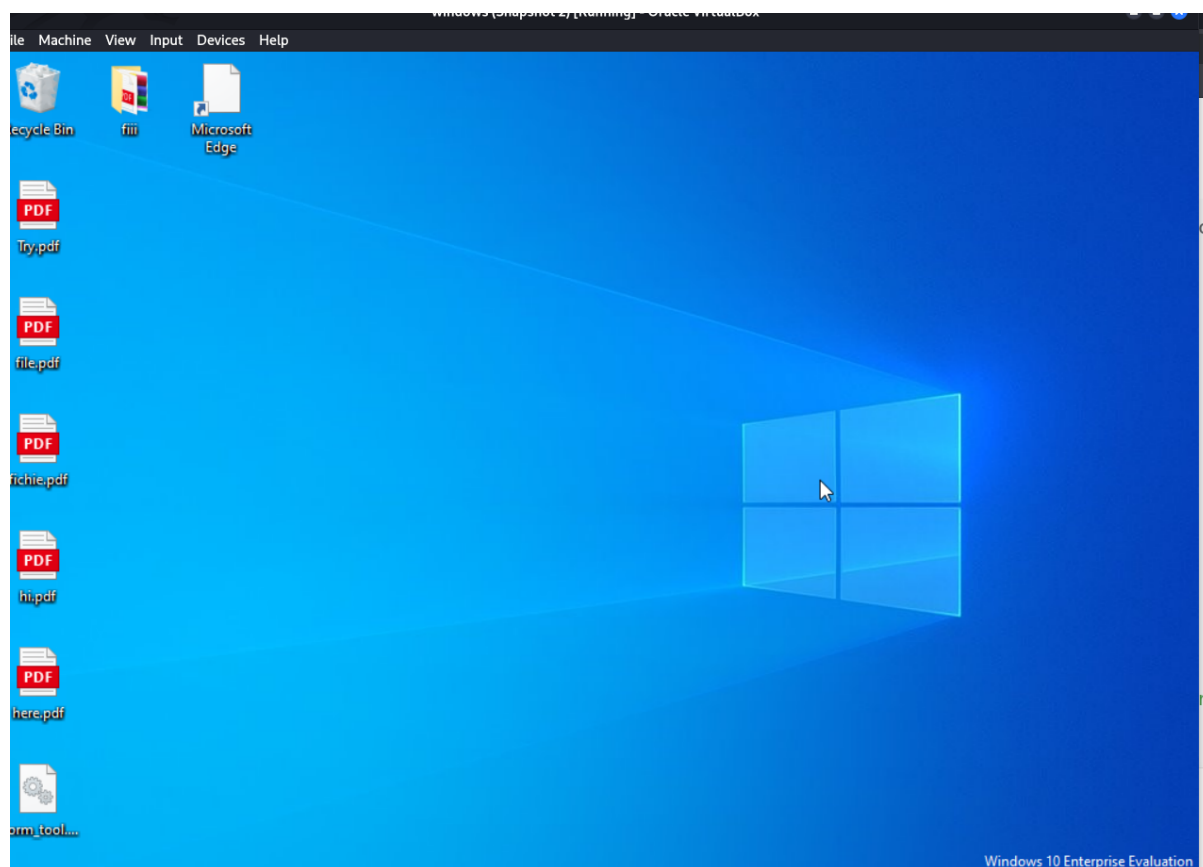


Figure 3: System restoration

5 Phase 4: Recovery

5.1 System Restoration

Method used:

- Restoring the VM to a previous clean snapshot, note that in real scenarios where the infected machine is a real PHYSICAL machine, we restore the system by reverting to restore points if not destroyed by malware, or in our case we need to reinstall the Windows OS and get back the files from backups, because the malware attempts to destroy the OS if rebooted or not paid the ransom.

5.2 Validation Steps

1. System Integrity Check

- After restoring the snapshot, no malicious process were found, meaning that the system is back to its clean state.

2. Network Connectivity

- No malicious traffic detected

3. Functionality Test

- Normal applications functioning
- User data accessible

5.3 Time to Recovery

- **Total Downtime:** 17 minutes and 38 seconds
- **Recovery Method:** Snapshot restoration

6 Phase 5: Lessons Learned

6.1 Incident Summary

What happened?

The user downloaded a suspicious file from a phishing website and executed the file. The malicious file dropped the process of the malware and started execution, it disabled the task manager and locked the screen, encrypted the files, and finally showed the ransom note. Rebooting or Powering off the system leads to fully crashing the OS.

6.2 Root Cause Analysis

Why did this happen?

- User executed unknown file downloaded from a suspicious website.
- Real-time detection of malwares and Virus analysis of Windows defenders was turned off.

6.3 What Worked Well

- Quick detection through Process Monitor
- VM snapshot enabled fast recovery
- Network isolation prevented lateral movement and any possibly hidden C2 connections
- Recovering the system via encryption key found in the decompiled code.

6.4 What Could Be Improved

- Automated alerting needed
- Response time could be faster
- Better user training required

7 Recommendations

7.1 Immediate Actions

1. Enable Real-Time Antivirus Protection

- Enabeling windows defenders real time detection and protection.

2. Network Segmentation

- Isolate critical systems from public network
- Implement VLAN segmentation

7.2 Long-Term Improvements

1. Regular Backups

- Use of regular backups to revert in case of any incident.
- Regular backup testing to confirm the integrity of the data.

2. User Security Awareness Training

- Sensibilation against phishing
- Educate on safe computing practices

3. Enhanced Monitoring

- Use of SIEM Solutions like Elastic Stack or Splunk for monitoring and logging
- Configure rules of detections like YARA Rules

8 Appendices

8.1 Appendix A: Technical Details

Attribute	Value
Malware Family	WormLocker 2.0
File Size	322.50 KB
File Type	PE32 executable
Compiler	Microsoft Visual C / .NET
Packer	Not Packed
Entropy	7.330003

Table 1: Malware Technical Attributes

8.2 Appendix B: IOC List

- **File Hashes:**

- MD5: 1b4f89bdb12a349de92ca7f1261e67a0
- SHA1: f368916850332757d7ed2f0ee335c16b9c9fc95b
- SHA256: d4c83205cf6f3098ab6a757312525f4d14a57a819306eeea5c0d022b00b38cf3

- **Network Indicators:**

- No network connection was established

- **Registry Keys:**

```
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System\DisableTaskMgr
HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Winlogon
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run
HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run
```

8.3 Appendix C: Tools Used

- SysMon
- Elastic
- Wireshark / tcpdump
- WinLogBeat
- Suricata
- Volatility Framework

- Autopsy
- VBoxManage
- Strings
- pescan
- monodis
- Detect it easy (die)
- upx packing tester
- YARA
- Dnspy
- Antivirus: Windows Defender

9 Report Sign-off

Prepared by: MADENE Meriem & MOSTEFAOUI Manel & BRAHMI Sarah Kaouther

MADENE Meriem
DFIR Responsible
Date: 14/11/2025

Reviewed by:

MR BENDELLA Mohammed
Professor of Network and Computer Security module
Date: